Tangible Compass
Haptic Navigation for Accessibility

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CRAFT ^2

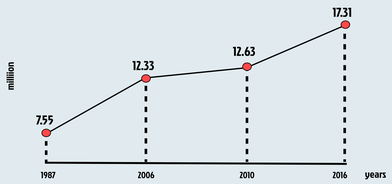
"Making navigation easier for people with visual impairments using haptics technology."

Problem: Challenges in Accessible Navigation

Many people rely on smartphones for navigation, but these devices mainly use visual and audio signals. For people who are blind or have low vision, visual feedback is not helpful, and audio cues can be missed in noisy environments. Existing solutions like high-tech navigational aids are often expensive and difficult to use. There's a need for an affordable, easy-to-build device that can provide reliable guidance through touch.

16.91 M visual impairments in China

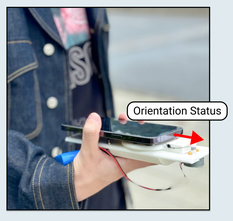
Increasing amount of visually impaired people in China over the years [1]

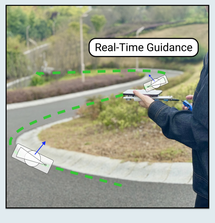


Solution: Tangible Compass

Tangible Compass is a handheld, DIY-friendly device that provides haptic feedback to guide users in real time. It connects to a smartphone, processes GPS and sensor data, and then uses a motor to deliver precise force-based cues that nudge the user in the correct direction. By translating direction into a physical sensation, users can navigate without needing to look at a screen or rely solely on hearing.

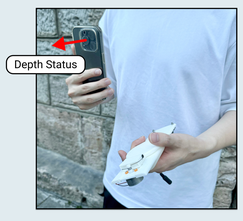
ⓐ Route Navigation

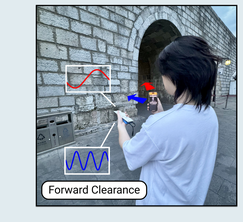


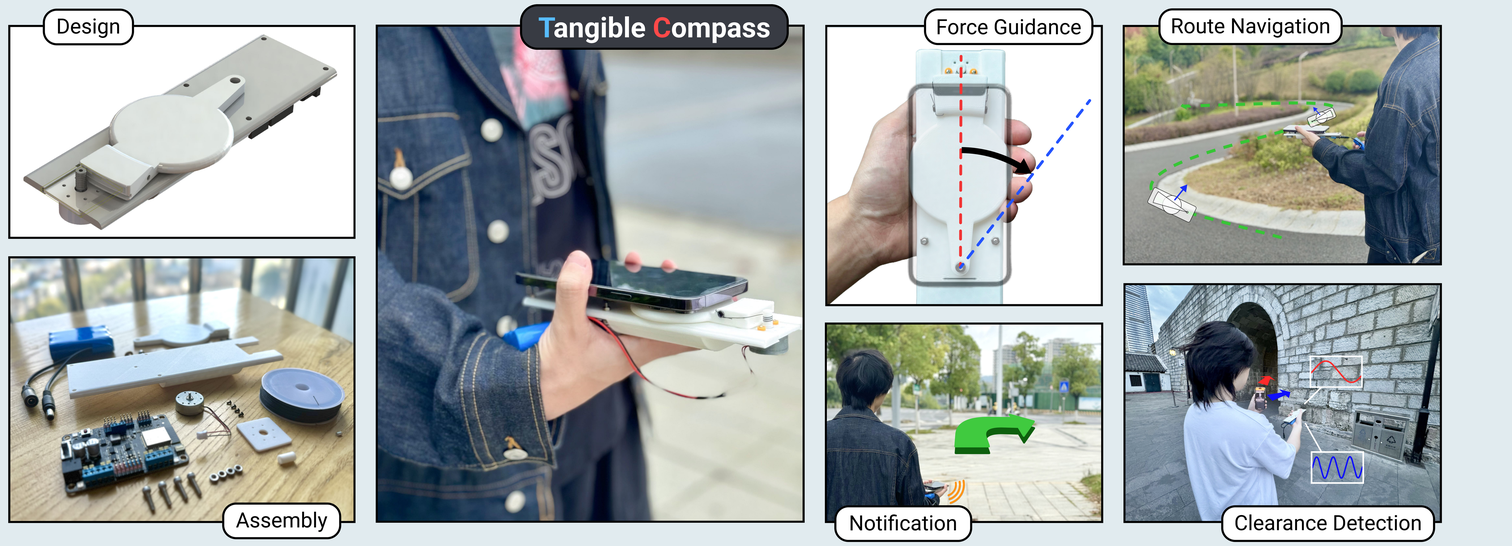


ⓑ Notification Procedure

ⓒ Clearance Detection







Key Features:

Affordable & DIY: Built using common materials and 3D-printed parts, it is accessible for makers and researchers alike.

Force-Based Feedback: The device gives gentle pushes in the hand to guide the user, providing a more intuitive and continuous form of navigation.

Multi-Scenario Use: Can be adapted for different environments—whether it's guiding someone along a route, notifying them of an upcoming turn, or detecting obstacles ahead.

How It Works: Haptic Feedback in Action

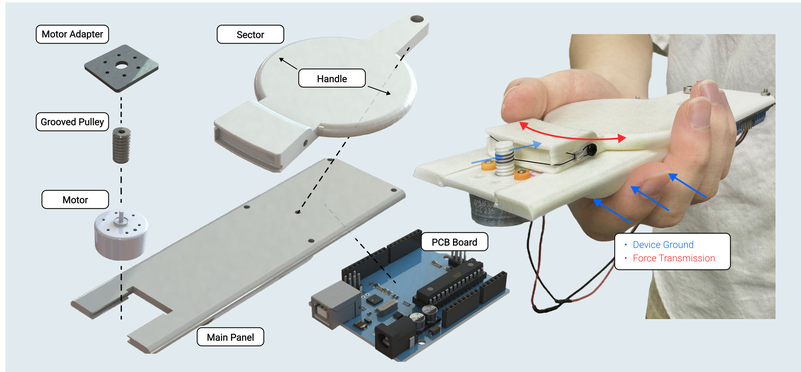
Tangible Compass implements a one-degree-of-freedom (1-DoF) kinesthetic system, meaning it provides feedback through controlled physical movement in a single direction. This feedback is driven by a small motor connected to the PCB. The motor pulls a cord that creates a force felt in the user's hand, guiding them by physically "pulling" them in the right direction.

Technical Insight:

The motor system uses a capstan drive, which ensures smooth and precise force transmission with minimal friction [2]. This allows for accurate, reliable feedback even in complex navigation scenarios.

With real-time data streaming from phyphox physics experiment app [3], Tangible Compass translates sensor information into directional force, ensuring that the user is always guided toward their destination.

A compact design and lightweight materials make the device portable and easy to use alongside any smartphone.



Future of Haptic Navigation: What's Next?

Tangible Compass shows how low-cost, open-source technology can improve the everyday lives of people with disabilities. Future iterations could include multi-directional force feedback, smaller wearable versions (e.g., wristbands), and more complex algorithms for detecting obstacles or changes in terrain. The ultimate goal is to make navigation not just accessible but intuitive for all users, regardless of their abilities.

References

[1] CDPF, CAPVD, DATAGOO
[2] Stanford University. Hapkit project.
[3] S. Staacks et al., "Advanced Tools for Smartphone-Based Experiments: phyphox," Physics Education, vol. 53, no. 4, p. 045009, 2018.